



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2017
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744

Paper 4 Practical

12 September 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams and graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

Suggested Answers

For examiner's Use	
1	/ 21
2	/ 20
3	/ 14
Total	/ 55

This paper consists of ____ printed pages.

[Turn over]

QUESTION 1

Fig. 1.1 shows the structure of a dormant seed that has been cut in half (longitudinal section).

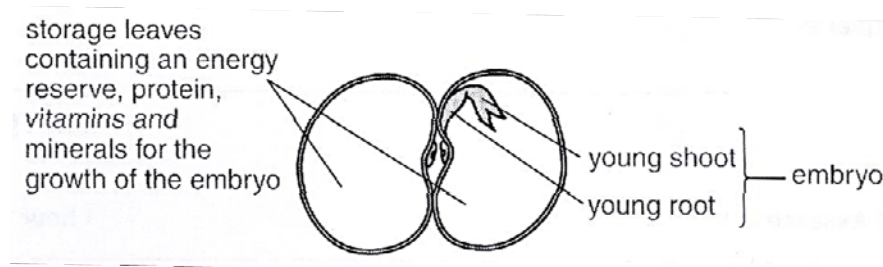


Fig. 1.1

Dormant seeds have a very low rate of respiration. When water is absorbed by dormant seeds, growth hormones are activated. These hormones activate genes that code for the synthesis of enzymes. These enzymes are used to hydrolyze the food reserves so they can be used for respiration and growth. The respiration rate can be measured using a respirometer.

Fig. 1.2 shows the respirometer.

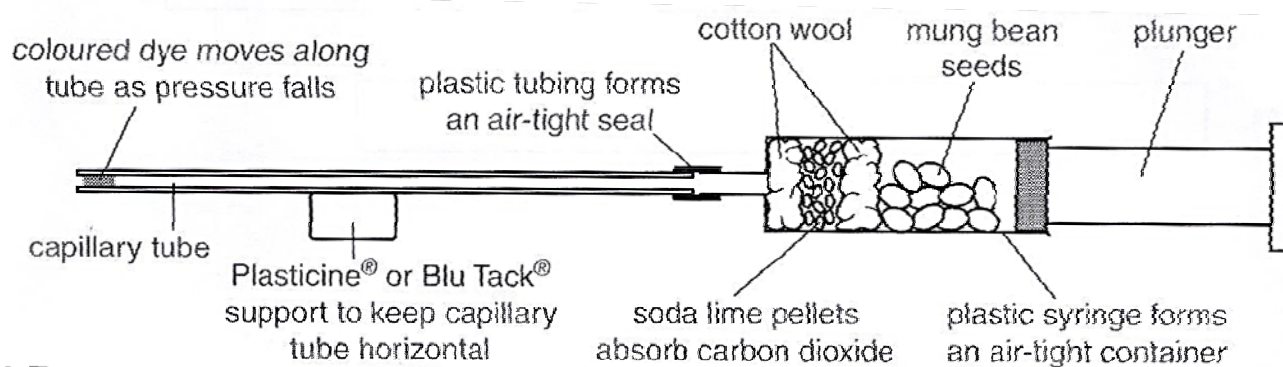


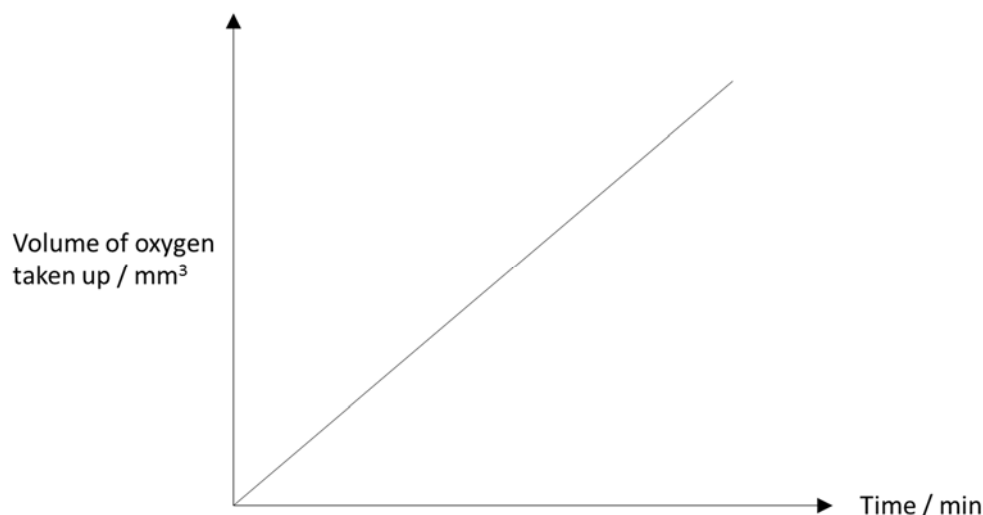
Fig. 1.2

As the seeds respire, oxygen is removed from the air and carbon dioxide is released. This carbon dioxide is absorbed by the soda lime. As the oxygen is used by the seeds, the pressure falls, causing the coloured dye to move along the capillary tube.

You are required to investigate the rate of oxygen uptake by respiring seeds.

Soda lime is harmful and corrosive. Safety glasses and gloves should be worn.

1. Sketch a fully-labelled graph to show the expected relationship between the volume of oxygen uptake and time. [1]



[1] Axes labelled correctly, straight line starts at origin (0,0), positive gradient, and does not plateau. Do not mark for absence of units

Proceed as follows:

Fig. 1.2 shows the respirometer that you will be setting up and using.

- Place a small plug of cotton wool in the respirometer and use a glass rod to gently push it to the bottom. The cotton wool must **NOT** be compacted.
- Using a spatula, add soda lime pellets on top of the cotton wool plug in the respirometer. The soda lime pellets should form a layer of about 1 cm deep.
- Add another small plug of cotton wool to the respirometer and gently push it down until it is just above the soda lime pellets. Do **NOT** compact the cotton wool.
- Take 10 mung beans and briefly dab them dry with paper towels. Find and record their mass.

Mass: 4.8 g

- Add the 10 mung bean seeds to the respirometer. Place the plunger. Leave the respirometer for 5 minutes.

Explain the purpose of leaving the respirometer for 5 minutes.

[1]

- Allow the soda lime to completely absorb/remove all CO₂ in the syringe chamber.

- Read through the remaining steps up to step **15** and decide on the results that you will be recording. Prepare a table to record all of these results in the space provided under step **15**.
- Place the end of the capillary tube into the dye solution and use the syringe plunger to pull in about 1 cm length of the dye. Wipe off any excess dye on the outside of the capillary tube.
- Place a ruler alongside the capillary tube and use a support, such as Plasticine® or Blu Tack®, to keep the capillary tube horizontal and aligned with the ruler.

10. Record the **cumulative distance** moved by the dye at every **30-second intervals** for **5 minutes**. Do not start the stopwatch until the dye has started to move.
11. If the dye has reached the end of the capillary tube before 5 minutes, pause the stopwatch but do not reset it to zero. Make a note of the cumulative distance moved by the dye up to this point. Reset the respirometer by pushing the plunger to move the coloured dye to the start position. As soon as the dye starts moving again, restart the stopwatch and continue recording the cumulative distance moved by the coloured dye at the completion of each 30-second interval, up to 5 minutes. These measurements will need to include both the cumulative distance noted when the respirometer was reset and the distance moved subsequently.
12. After 5 minutes, carefully expel the coloured dye onto a piece of paper towel by pushing in the plunger.
13. Carefully pull the plunger out of the syringe completely, without disturbing the contents. Replace the plunger and leave for 5 minutes. While waiting, proceed to step 16.
14. After 5 minutes, repeat steps 8 to 11 to measure and record a second set of results for these seeds.
15. Record all of your results in the table you have prepared below. [5]

Teacher's trial results on 06/10/2016 for reading 1

	Cumulative distance travelled by the coloured dye / mm		
Time interval / s	Reading 1	Reading 2	Average
30	17	19	26.5 or 27
60	36	34	35.0 or 35
90	61	60	60.5 or 61
120	74	81	77.5 or 78
150	93	90	91.5 or 92
180	110	117	113.5 or 114
210	126	127	126.5 or 127
240	140	143	141.5 or 142
270	150	151	150.5 or 151
300	166	163	164.5 or 165

[1] Correct heading for independent variable: in first column, time in s or in min.

[1] Correct heading for dependent variable: distance in either mm or cm.

[1] Records 2 readings, each comprises 10 intervals

[1] Readings recorded to nearest mm or 0.5mm or 3 s.f. / nearest 0.05cm or 0.1cm or 3 s.f.

[1] Correctly calculate average reading, in nearest mm or 0.5mm or 3 s.f. / nearest 0.05cm or 0.1cm or 3 s.f.

16. Suggest an advantage of recording the cumulative distance moved by the coloured dye every 30 seconds, instead of only recording the total distance moved after 5 minutes. [1]
 - **Ref to.** Checking that the rate is constant/steady

17. Assuming that a 10 mm length of capillary tubing has a volume of 8.0 mm^3 , calculate the mean rate of oxygen consumption of the mung bean seeds per gram of tissue in $\text{mm}^3\text{s}^{-1}\text{g}^{-1}$ over the entire 5 minutes. [3]

Show **all** the steps in your calculation, including relevant units at each step.

Volume of oxygen consumed
= $(8.0 \text{ mm}^3/10\text{mm}) \times 165\text{mm}$
= 132.0 mm^3 [1]

Rate
= $132.0 \text{ mm}^3 / 4.8\text{g} / 300\text{s}$ [1]
= $0.0917 \text{ mm}^3\text{s}^{-1}\text{g}^{-1}$ [1]

[1] calculation of volume from the distance travelled by the dye

[1] dividing volume by mass of seeds **and** time (**marking for working**)

[1] calculation of mean rate of oxygen consumption **and** recorded to 3 s.f. (**allow e.c.f.**)

Mean rate of oxygen consumption: 0.0917 $\text{mm}^3\text{s}^{-1}\text{g}^{-1}$

18. Describe a control for this experiment and explain the rationale for this control. [2]

- Replace soaked seeds with seeds that have not been soaked in water
- To show that only germinating seeds undergo respiration to take up oxygen

OR

- Replace soaked seeds with boiled seeds.
- To show that only germinating seeds undergo respiration to take up oxygen

OR

- Replace soaked seeds with glass beads.
- To show that only germinating seeds undergo respiration to take up oxygen

19. Table 1.1 shows the results from an experiment to measure the rate of oxygen consumption of 15 pea seeds at different temperatures. The experiment was repeated three times for each temperature, and the average rate was calculated.

Table 1.1

Temperature / °C	Average rate of oxygen consumption / mm ³ s ⁻¹	Average rate of oxygen consumption per gram of tissue / mm ³ s ⁻¹ g ⁻¹
10	8.30	<u>11.1</u>
15	11.0	<u>14.7</u>
20	15.8	<u>21.1</u>
25	20.0	<u>26.7</u>
30	33.5	<u>44.7</u>

- (a) The mean mass of one pea seed is 50.0 mg.

Complete Table 1.1 by calculating the average rate of oxygen consumption **per gram** of tissue in mm³s⁻¹g⁻¹.

Show your working for the result at 10°C, in the space below. [2]

[1] for calculation

- Mass of one seed = 50 mg = 0.05 g
- Mass of 15 seeds = 0.75 g
- Rate = 8.3 mm³s⁻¹ / 0.75g = 11.0666 mm³s⁻¹g⁻¹ = 11.1 mm³s⁻¹g⁻¹

[1] for completing table to 3 s.f. (e.c.f. for not converting to grams)

- (b) State and explain the most important variable that needs to remain constant throughout the experiment. [2]

- Number of seeds
- More seeds → more oxygen taken in → will not be a true reflection of the effect of temperature

- (c) The average rate of oxygen consumption was compared between 10°C and 15°C using Student's t-test. The calculated t-value was determined to be 1.740.

Degrees of freedom	Significance level					
	0.50	0.20	0.10	0.05	0.02	0.01
1	1.000	3.078	6.314	12.706	31.821	63.657
2	0.816	1.886	2.920	4.303	6.965	9.925
3	0.765	1.638	2.353	3.182	4.541	5.841
4	0.741	1.533	2.132	2.776	3.747	4.604
5	0.727	1.476	2.015	2.571	3.365	4.032
6	0.718	1.440	1.943	2.447	3.143	3.707

Using the t-distribution table above, explain what conclusions can be drawn from the calculated t-value. [3]

- At $df = 4$, calculated t-value is smaller than the critical t-value of 2.776 at 0.05 significance level.
- The probability that the difference between the rate of oxygen consumption at 10°C and 15°C being due to chance is between 0.1 to 0.2, which is more than 0.05 (cut-off).
- The difference is hence not significant and is due to chance.
- **Idea that** the increase in temperature from 10°C to 15°C did not affect the average rate of oxygen consumption.

- (d) In a further investigation, using the same respirometer at 20°C, the soda lime was removed and the experiment repeated. The dye did not move.

Suggest why the dye did not move. [1]

- **Idea that** volume of oxygen uptake equals volume of carbon dioxide release, therefore no change in overall volume in the respirometer / air pressure unchanged

[Total: 21]

QUESTION 2

Urea, **U**, reacts with water to form aqueous ammonium carbonate. Aqueous ammonium carbonate produces ammonium ions. These form an alkaline solution which causes red litmus paper to turn blue. The time taken for red litmus paper to turn blue can be used to monitor the progress of the reaction.

K is known to play a role in the above reaction. You are required to investigate the effect of concentration of solution **K** on this reaction.

You are provided with:

- 25 cm³ of 10.0%, **K**, which is an irritant.
- 100 cm³ of distilled water, **W**.
- 25 cm³ of a solution of urea, **U**.
- Red litmus paper, each about 6 cm in length.

It is recommended that you wear safety goggles.

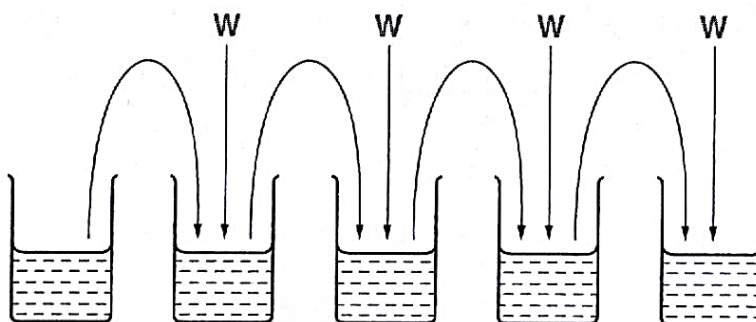
1. Carry out **serial** dilution of solution **K**, to reduce the concentration of the solution by half between each of four successive dilutions, and set up a control.

Label four small beakers, **D1**, **D2**, **D3** and **D4**, for the serial dilutions, and label another small beaker, **C**, for the control.

Complete the table below to show how you will make the different concentrations of solution **K** and how you will set up the control, **C**.

Record the values to **3 significant figures**.

[2]



Label	K	D1	D2	D3	D4
Concentration of K / %	10.0	5.00	2.50	1.25	0.625
Volume of solution K taken from the previous dilution / cm ³		10.0	10.0	10.0	10.0
Volume of distilled water, W / cm ³		10.0	10.0	10.0	10.0

Description of the control, **C**:

- Replace solution **K** with 10 cm³ distilled water, i.e. 10 cm³ distilled water + 10 cm³ distilled water

[1] correct concentrations to 3 s.f., + correct volume in dilution table

[1] description of control with appropriate volumes quoted [**Reject**: equal volume]

2. In order to monitor the progress of the reaction, in step 4, red litmus paper will be added to each mixture of solution **K** and solution **U**, in a test-tube. To prevent the paper from sticking to the wall of the test-tube, you will need to use the glass rod to add it as follows:

Cut a piece of red litmus paper so that it is a little shorter than the circumference of the glass rod. Moisten the paper and stick it to the end of the glass rod as shown in Fig. 2.1. The glass rod can then be lowered into the mixture of **K** and **U**. The red litmus paper will slip off into the mixture and the glass rod can then be removed.

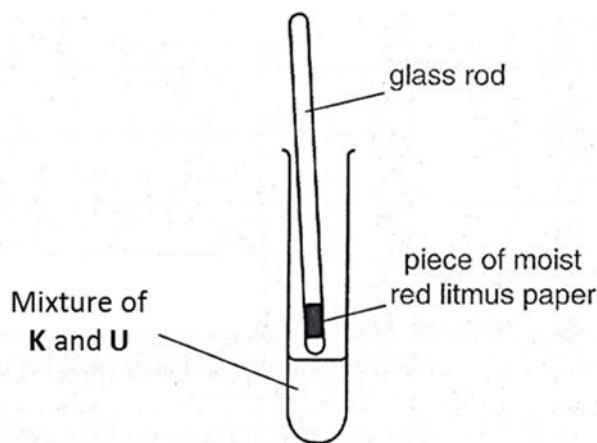


Fig. 2.1

Proceed as follows:

3. To test the activity of the highest concentration of solution **K**, put 2 cm³ of **U** into a test-tube, then add 2 cm³ of **K** and mix well. The reaction will start as soon as **K** is added.

Immediately put one piece of red litmus paper into the test-tube as described in step 2 and start timing.

4. Record the time taken for the piece of red litmus paper to turn blue. If the piece of red litmus paper does not turn blue in 10 minutes, record as 'more than 600'.
5. Repeat steps 3 and 4 for the other concentrations of solution **K**, and the control, **C**. The red litmus paper used each time should be of the same size.
6. Using an appropriate format, record the results of this investigation for the various concentrations of solution **K**, including the control, in the space provided (in step 7).

7. Use the space below to record your results.

[3]

Teacher's trial on 18/10/2016

Concentration of solution K / %	Time taken for red litmus paper to turn blue / s
10.0	29.1
5.00	40.3
2.50	54.4
1.25	71.4
0.625	102.5
0.00	More than 600

- Correct headings with units [Reject: D1, D2, D3, D4]
- Complete data (from 0%-10% urease) + correct trend (shortest time for highest conc., longest time for lowest conc., 'more than 600' for control)
- Values recorded to appropriate precision of the stopwatch (i.e. 1 d.p.)

8. From the results of your investigation, suggest the identity of K.

[2]

- K is urease.
- K increases the rate of reaction → time taken for ammonium carbonate to be produced to turn red litmus paper blue decreases from 102.5s to 29.1s when (enzyme) concentration of solution K increases from 0.625% to 10.0%.

9. Calculate the rate of reaction using your result for 10.0% concentration of solution K. Show your workings clearly. [1]

$$\begin{aligned}\text{Rate} &= 1 / \text{time} \\ &= 1 / 29.1\text{s} \\ &= 0.0344 \text{ s}^{-1} \text{ (3 s.f.)}\end{aligned}$$

[1] Rate recorded to 3 s.f.

Rate of reaction: s⁻¹

10. (a) Lack of replicates is a limitation of this procedure.

Describe one other limitation.

[1]

- Difficulty in judging when litmus paper changes colour from red to blue

(b) Suggest how you would make one improvement to this procedure to reduce the effect of the significant source of error identified in **10(a)**. [1]

- Use a pH meter to measure the time taken for pH to reach a particular pH

11. The effect of pH on the activity of two proteolytic enzymes, **A** and **B**, was compared. The substrate of the enzymes was coloured jelly, which is made of proteins. It is known that both enzymes work best at 38°C.

The apparatus for each pH was set up as shown in Fig. 2.2.

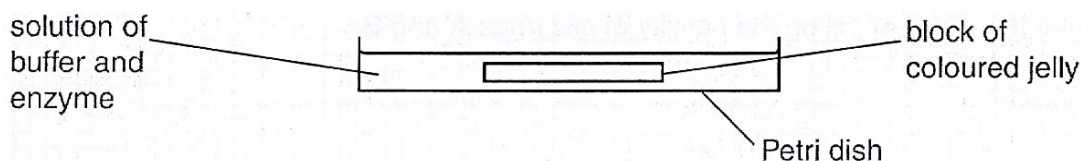


Fig. 2.2

The block of coloured jelly gets smaller as it is digested by the enzymes.

- (a) State two variables which would need to be controlled and suggest how each variable would be controlled. [2]

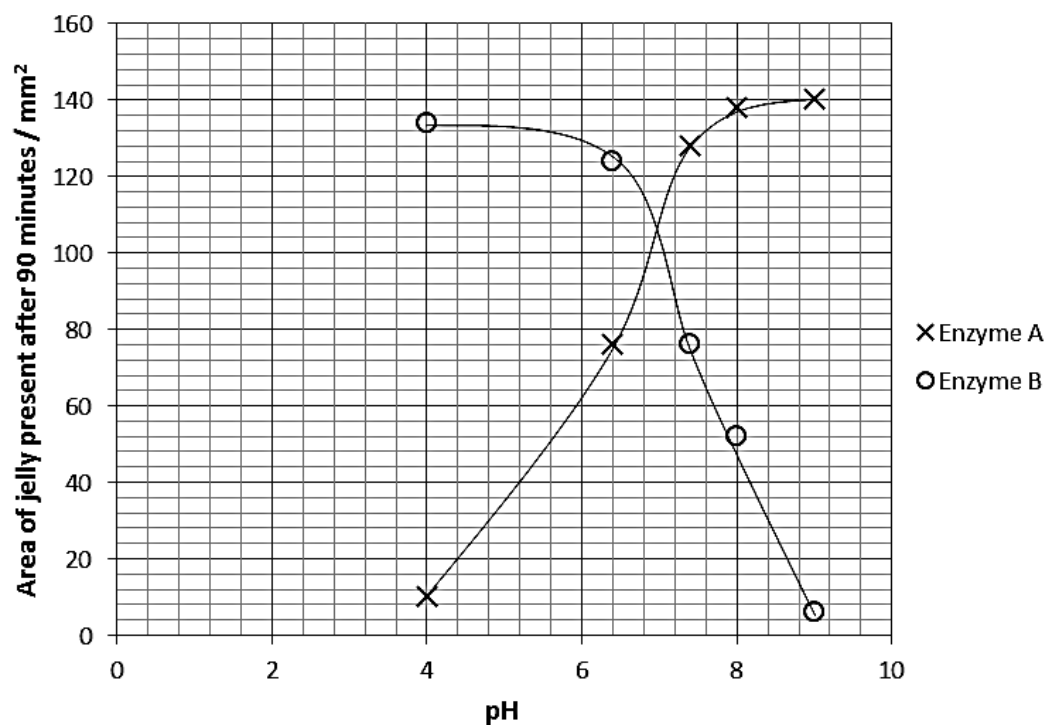
- **Dimensions / Size** of jelly block at the start of experiment. Measure exact **length** and **width** using **ruler** and knife.
- **Concentration** of **enzymes** A and B. (*Idea that*) Make a master concentration of enzyme A and B to be used throughout the experiment.
- **Volume** of **buffer** **OR** **enzymes** A and B. Use **syringe/ measuring cylinder/pipette** to ensure consistent volume.
- **Temperature** at 38°C. Use a thermostatically-controlled **incubator**.

The results of the investigation are shown in Table 2.1.

pH	Area of jelly present after 90 minutes / mm ²	
	enzyme A	enzyme B
4.0	10	134
6.4	76	124
7.4	128	76
8.0	138	52
9.0	140	6

Table 2.1

- (b) Plot, on the grid below, the data shown in Table 2.1. Draw lines of best fit for enzyme **A** and enzyme **B**. [3]



- [1] No awkward scale (e.g. 3:10) AND graph occupies at least half the graph paper AND Points joined by smooth line of best fit
- [1] Correct choice of axes AND complete with labels & unites
- [1] Data plotted accurately without extrapolation AND key/label to distinguish the 2 sets of data (plotted with a x or dot with O)

(c) Describe the effect of pH on the activity of enzymes **A** and **B**. [1]

- For enzyme A, as pH increases from 4.0 to 9.0, the area of jelly present after 90 minutes increases from 10 to 140 mm², while for enzyme B, decreases from 134 to 6 mm²,

(d) Explain why changes in pH affect the activity of these two enzymes differently. [4]

- *(idea that)* Enzyme **A** works best in alkaline **pH 9**, enzyme **B** in acidic **pH 4**.
- Any changes in pH from their optimum, changes amount of protons/H⁺.
- Neutralization of charged and polar R groups, leading to breaking of ionic and hydrogen bonds that holds the enzyme in its 3D conformation
- Loss of 3D conformation of the enzymes, hence loss of specific shape of active site
- Active site no longer complementary to substrate, reduce enzyme-substrate complexes form, enzyme activity decreases.

OR

- Neutralization of charged and polar R groups at the active site, hence affecting the formation of ionic and hydrogen bonds between the enzyme and substrate.
- Reduce enzyme-substrate complexes form, enzyme activity decreases.

[Total: 20]

QUESTION 3 – Planning Question

You are required to plan an investigation to find out the effect of surface area-to-volume ratio on the rate of diffusion of hydrochloric acid into agar blocks containing phenolphthalein.

Phenolphthalein is an indicator which appears pink at pH higher than 7, and colourless at pH less than 7.

You must use:



- Sixteen 2cm x 2cm x 2cm phenolphthalein-containing agar blocks at pH 8
- 10 g dm⁻³ hydrochloric acid

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- white tile
- scalpel
- 15cm ruler

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of apparatus used.
- identify the independent and dependent variables.
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible.
- include the layout of result tables and graphs with clear headings and labels.
- use the correct technical and scientific terms.
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

[Background]

Simple diffusion of hydrochloric acid is a process where hydrochloric acid moves down a concentration gradient without the input of energy. It is affected by the surface area through which diffusion can occur.

Description of (simple) diffusion [1]

[Hypothesis]

The larger the surface area to volume ratio, the faster the hydrochloric acid will diffuse through the agar block, taking less time for the agar block to turn from pink to colourless [accept: white]

Correct hypothesis [1]

[Independent and dependent variables]

Independent variable: surface area to volume ratio (no units)

Dependent variable: time taken for agar blocks/phenolphthalein to turn colourless / s
[Accept min]

Both independent and dependent variables [1]

[Variables to be controlled]

- Total volume of agar blocks for each surface area-to-volume ratio obtained
- Concentration of hydrochloric acid
- Volume of hydrochloric acid
- Temperature
- Initial pH of agar blocks

Rationale:

Variables are kept controlled as these variables affect the rate of diffusion and/or time taken for the agar blocks to decolourise and hence can affect the accuracy of each independent variable examined.

At least 2 variables + rationale [1]

[Methods]

1. Using the scalpel and white tile, prepare the agar blocks of 5 different surface area-to-volume ratios according to the table below from the given 2cm x 2cm x 2cm agar blocks.

[Keeping total volume constant]

Number of blocks	Dimensions of each block /cm	Total surface area /cm ²	Total volume /cm ³	SA:V ratio
1	2 x 2 x 2	24	8.0	3:1
2	2 x 2 x 1	16 x 2 = 32	4.0 x 2 = 8.0	4:1
4	2 x 1 x 1	10 x 4 = 40	2.0 x 4 = 8.0	5:1
8	1 x 1 x 1	6 x 8 = 48	1.0 x 8 = 8.0	6:1
16	1 x 1 x 0.5	4 x 16 = 64	0.5 x 16 = 8.0	8:1

OR [per block]

Dimensions /cm	Surface area /cm ²	Volume /cm ³	SA:V ratio
2 x 2 x 2	24	8.0	3:1
2 x 2 x 1	16	4.0	4:1
2 x 1 x 1	10	2.0	5:1
1 x 1 x 1	6	1.0	6:1
1 x 1 x 0.5	4	0.5	8:1

2. Place the sixteen 1x1x0.5cm blocks into a 50cm³ [accept 100, 500cm³] beaker.
3. Ensure that all agar blocks do not overlap each other.
4. The starting colour of the agar blocks should be pink.
5. Fill the beaker with 20cm³ of 10 g dm⁻³ hydrochloric acid. The hydrochloric acid solution should cover all the blocks in the beaker.
6. Start the stopwatch.
7. As soon as all the blocks in the beaker turn colourless, stop the stopwatch and record the time taken.
8. Repeat steps 2-7 for the other agar block dimensions.
9. Set up a control beaker by replacing hydrochloric acid with distilled water for any of the block dimensions.
Rationale: This is to ensure that the colour change from pink to white in the agar blocks is due to hydrochloric acid diffusing into the agar blocks and no other factors.
10. Repeat steps 2-8 another two times to obtain 3 replicates. Calculate the average time taken to minimize error.
11. Repeat the entire experiment to ensure reproducibility of the trend obtained.

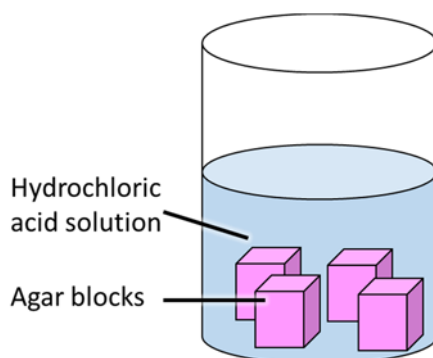
Any 3:

Keeping total volume of blocks in one beaker constant [1]
 Using an appropriate and constant volume of HCl AND using appropriate vessel [1]
 State that agar blocks should not overlap each other AND should be fully submerged [1]
 State that the initial colour of blocks is pink AND using a stopwatch to measure the time taken for blocks to decolourise [1]

Compulsory:

Table with appropriate block dimensions + 5 different SA:V ratios [1]
 Control and rationale [1]
 Replicates and repeats of the experiment + rationales [1]

[Experimental set-up]



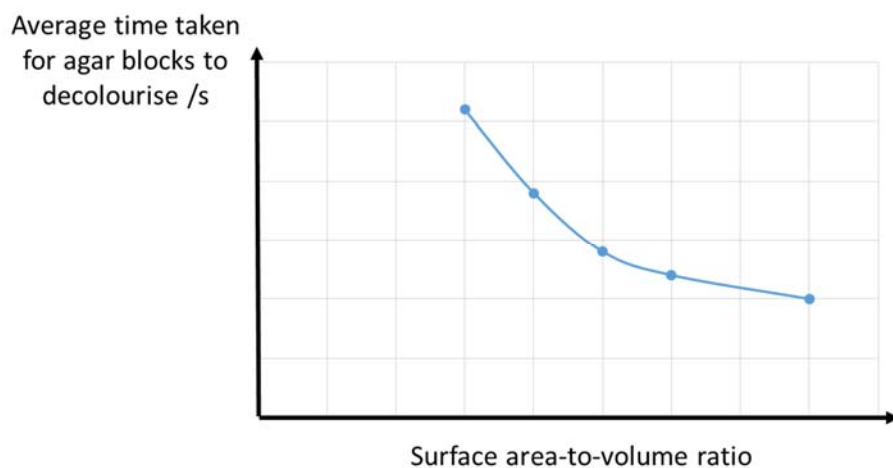
Labelled diagram of the set-up [1]

[Results – table]

Surface area-to-volume ratio	Time taken for agar blocks to turn colourless / s			
	R1	R2	R3	Average
3:1				
4:1				
5:1				
6:1				
8:1				

Correct headings with units; all SA:V ratios included [1]

[Results – graph]



Correct graph trend [1]

[Risks & Precautions]

Risk	Precaution
Hydrochloric acid is an irritant / corrosive to the skin/eyes	Wear latex gloves/ goggles. Wash off with water immediately if it comes into contact with skin.
Phenolphthalein in the agar blocks is carcinogenic	Wear latex gloves
Scalpel is sharp and may cut hands	Handle with care / wear thick gloves / use a blunt knife

Risk and corresponding precaution [1]